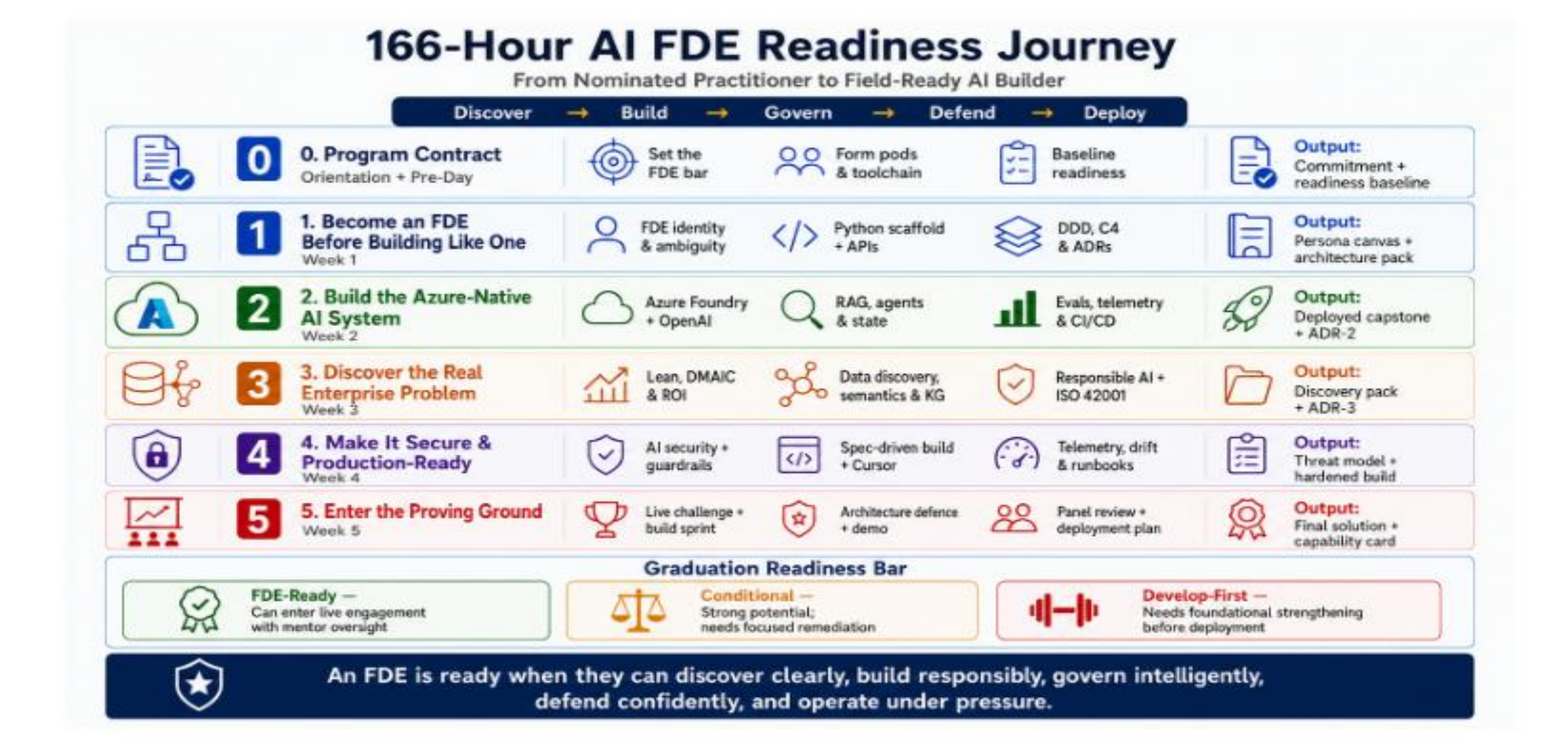
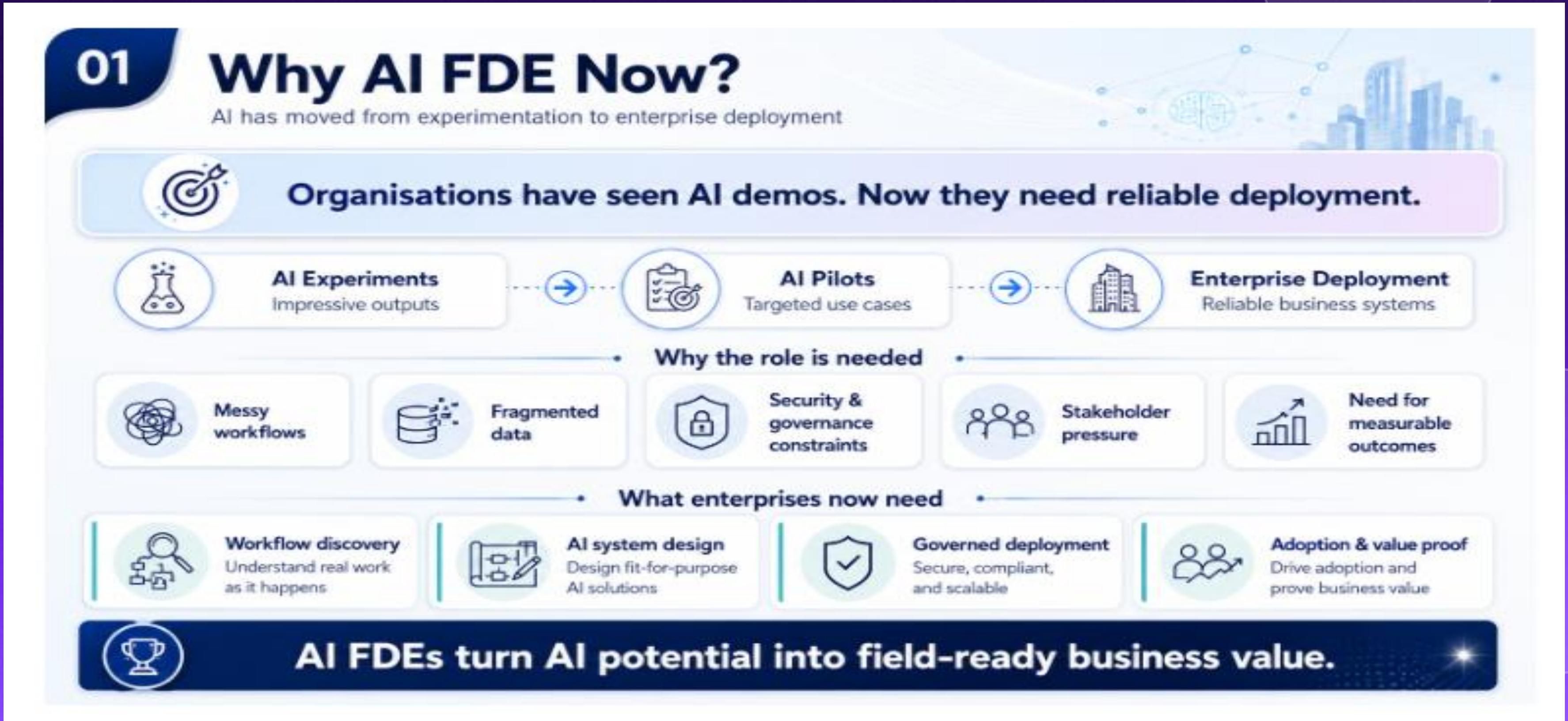

166-Hour AI FDE Readiness Journey
From Nominated Practitioner to Field-Ready AI Builder
Discover → Build → Govern → Defend → Deploy
0
0. Program Contract
Orientation + Pre-Day
Set the FDE bar
Form pods & toolchain
Baseline readiness
Output: Commitment + readiness baseline
1
1. Become an FDE Before Building Like One
Week 1
FDE identity & ambiguity
Python scaffold + APIs
DDD, C4 & ADRs
Output: Persona canvas + architecture pack
2
2. Build the Azure-Native AI System
Week 2
Azure Foundry + OpenAI
RAG, agents & state
Evals, telemetry & CI/CD
Output: Deployed capstone + ADR-2
3
3. Discover the Real Enterprise Problem
Week 3
Lean, DMAIC & ROI
Data discovery, semantics & KG
Responsible AI + ISO 42001
Output: Discovery pack + ADR-3
4
4. Make It Secure & Production-Ready
Week 4
AI security + guardrails
Spec-driven build + Cursor
Telemetry, drift & runbooks
Output: Threat model + hardened build
5
5. Enter the Proving Ground
Week 5
Live challenge + build sprint
Architecture defence + demo
Panel review + deployment plan
Output: Final solution + capability card
Graduation Readiness Bar
FDE-Ready — Can enter live engagement with mentor oversight
Conditional — Strong potential; needs focused remediation
Develop-First — Needs foundational strengthening before deployment
An FDE is ready when they can discover clearly, build responsibly, govern intelligently, defend confidently, and operate under pressure.


01
Why AI FDE Now?
AI has moved from experimentation to enterprise deployment
Organisations have seen AI demos. Now they need reliable deployment.
AI Experiments
Impressive outputs
AI Pilots
Targeted use cases
Enterprise Deployment
Reliable business systems
Why the role is needed
Messy workflows
Fragmented data
Security & governance constraints
Stakeholder pressure
Need for measurable outcomes
What enterprises now need
Workflow discovery
Understand real work as it happens
AI system design
Design fit-for-purpose AI solutions
Governed deployment
Secure, compliant, and scalable
Adoption & value proof
Drive adoption and prove business value
AI FDEs turn AI potential into field-ready business value.

02 What Is An AI FDE?

A field-deployed builder who converts unclear business problems into working, governed AI systems

An AI Forward Deployed Engineer works close to the customer, business unit, or operational team to turn ambiguity into deployable AI solutions.

The role combines

- Solution architect
- AI engineer
- Product thinker
- Domain translator
- Workflow analyst
- Governance & stakeholder operator

What makes the role different

- Works in the field
- Builds with context
- Owns adoption

The AI FDE owns the journey from ambiguity to adoption.

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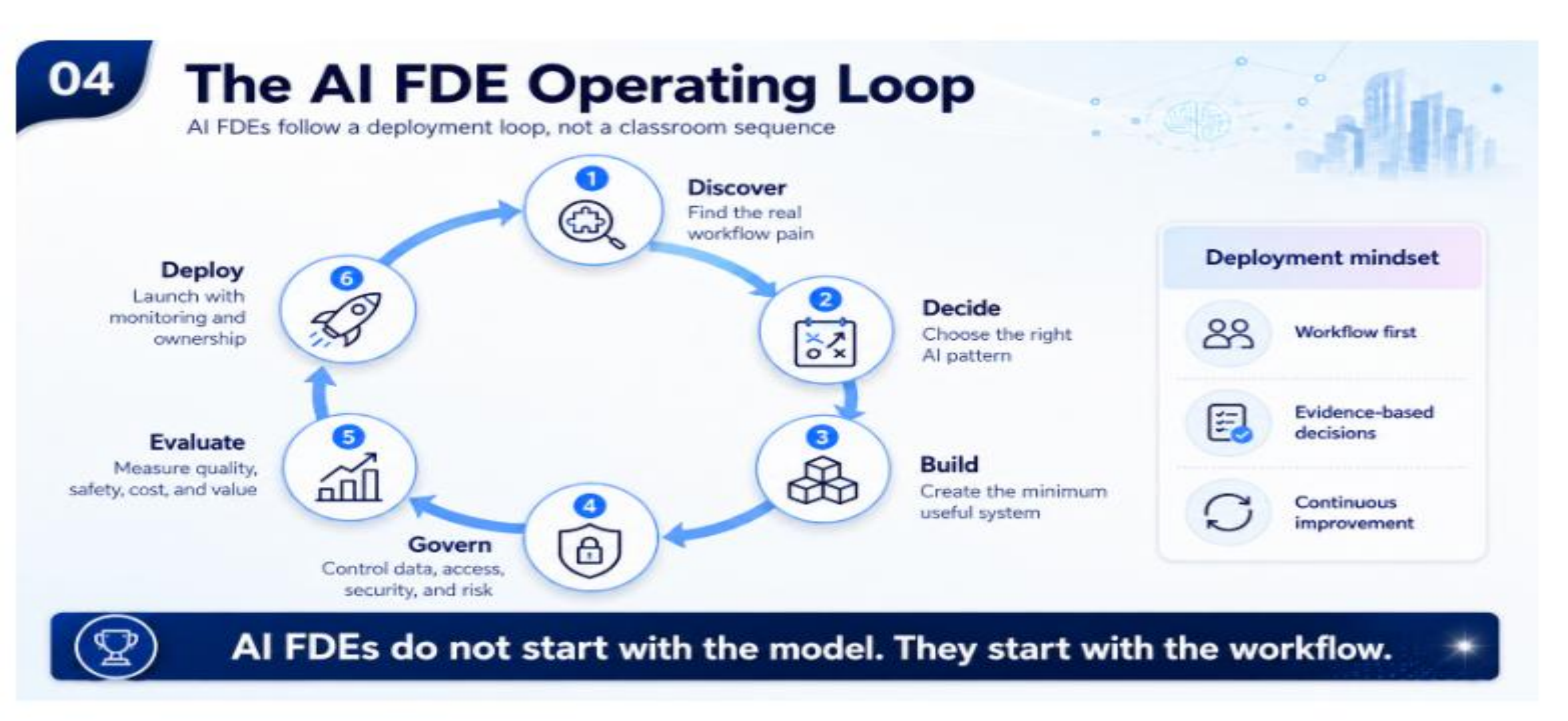
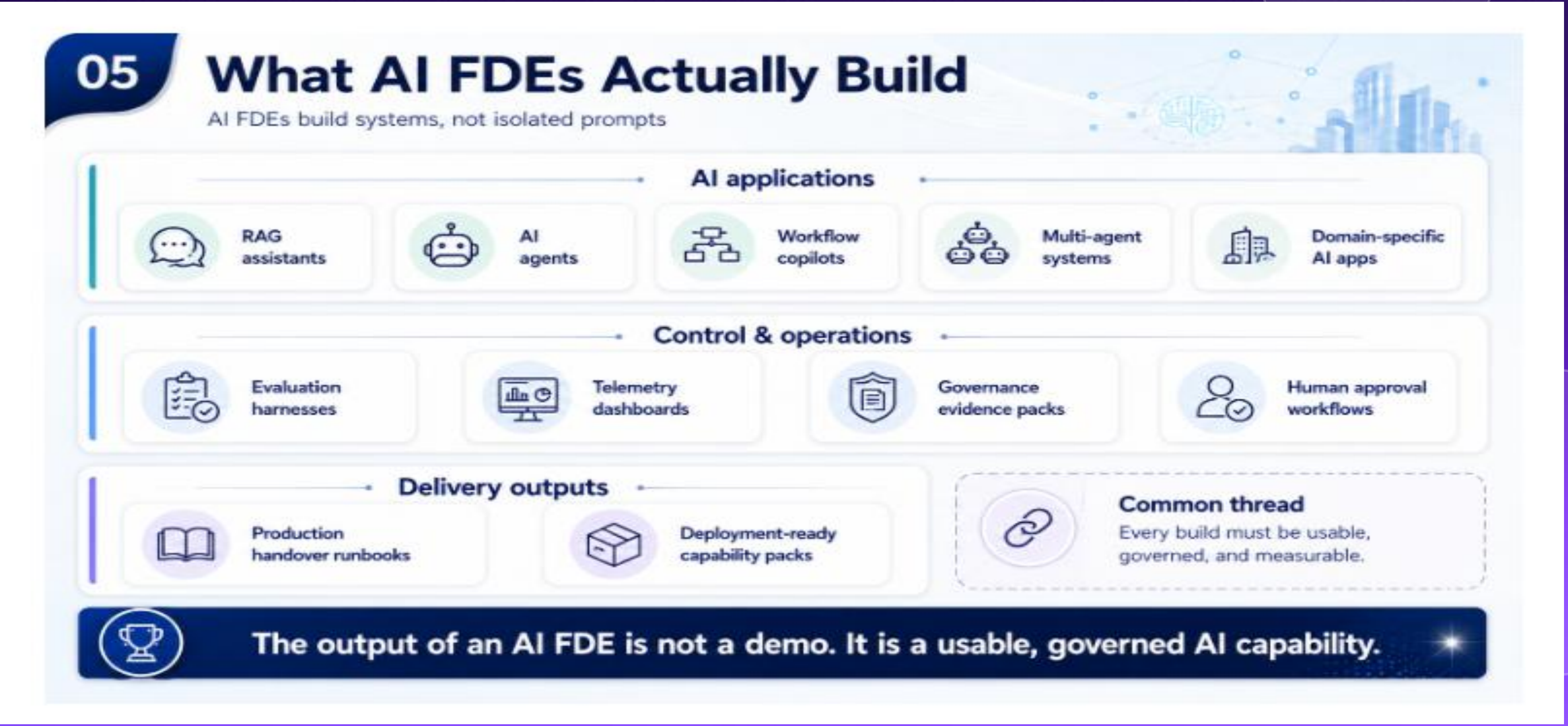
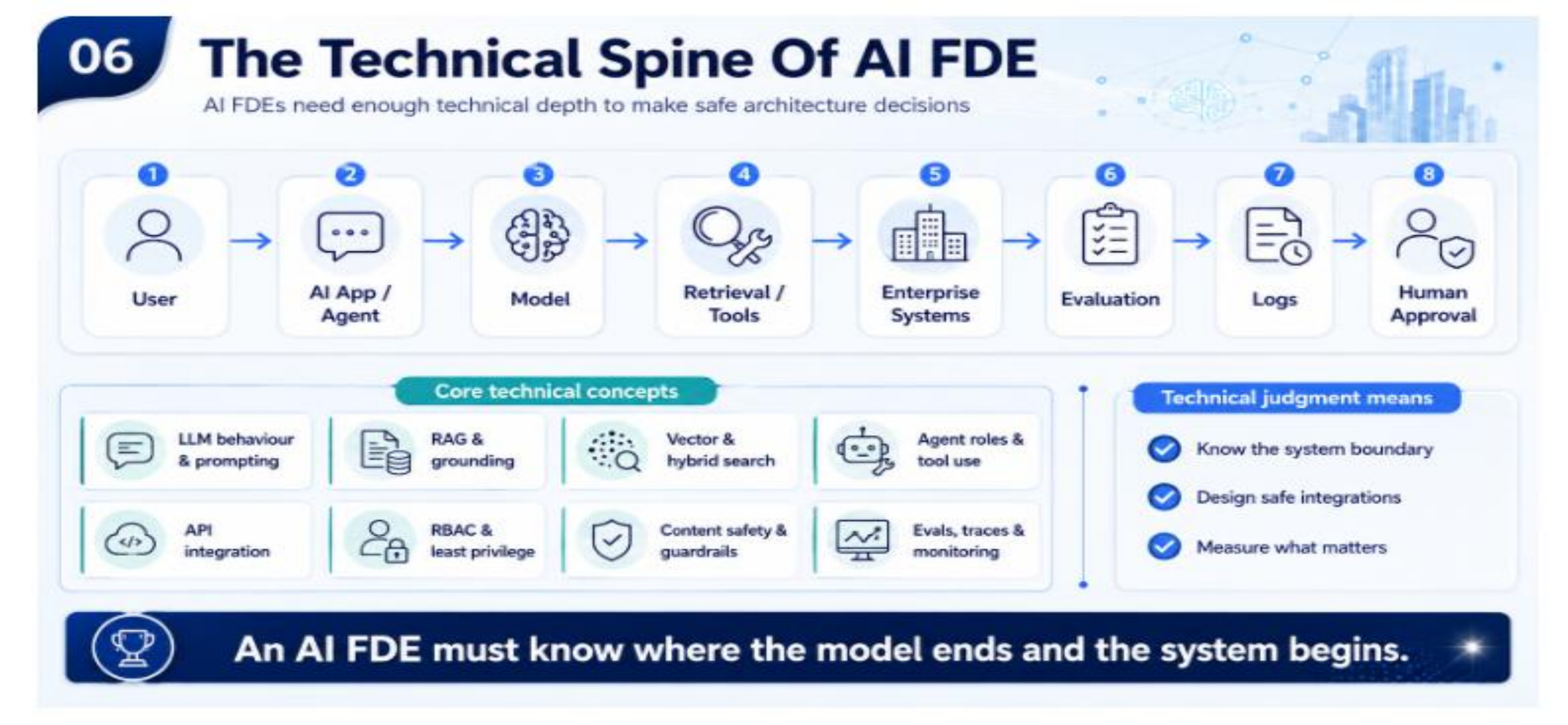
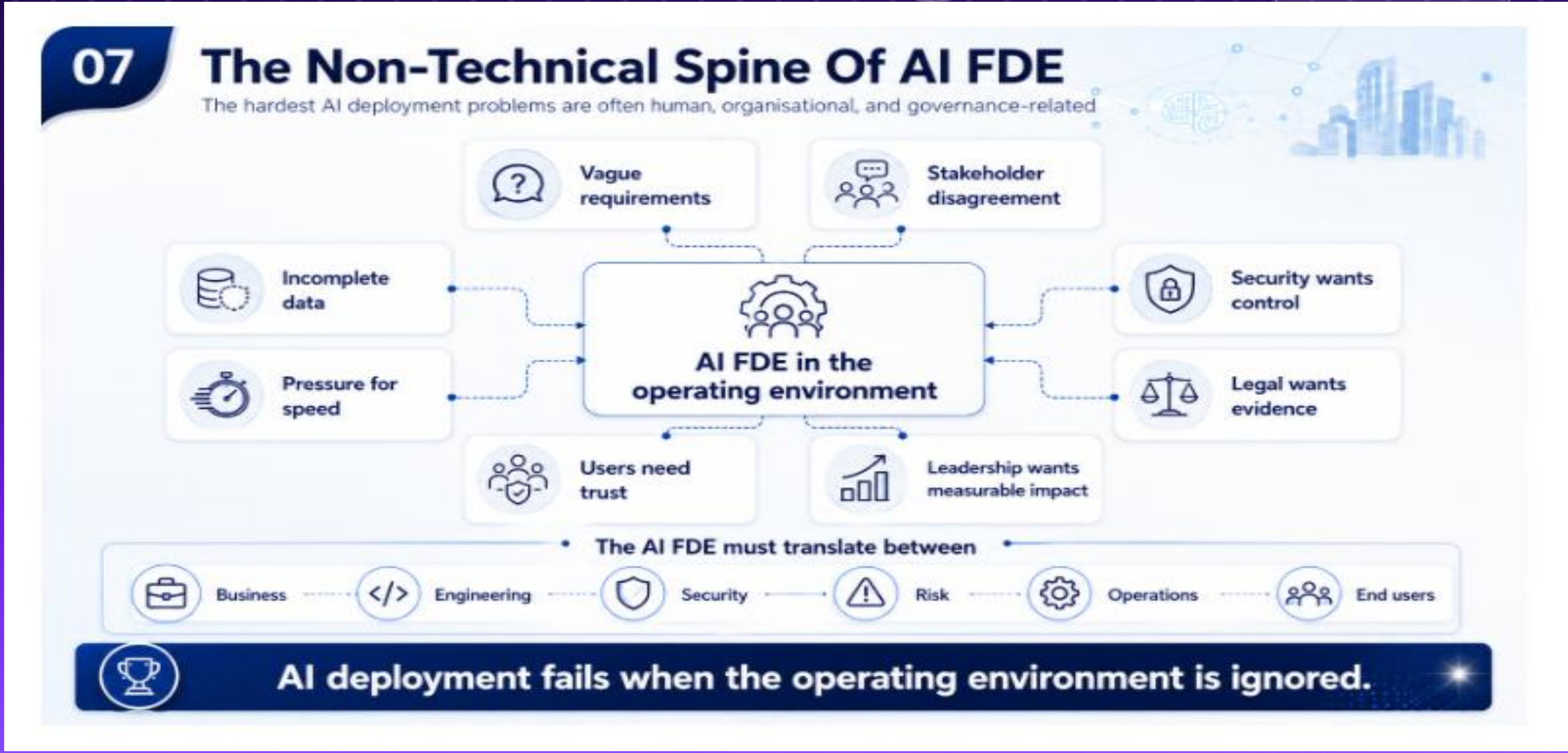

04
The AI FDE Operating Loop
AI FDEs follow a deployment loop, not a classroom sequence
1
Discover
Find the real workflow pain
2
Decide
Choose the right AI pattern
3
Build
Create the minimum useful system
4
Govern
Control data, access, security, and risk
5
Evaluate
Measure quality, safety, cost, and value
6
Deploy
Launch with monitoring and ownership
Deployment mindset
Workflow first
Evidence-based decisions
Continuous improvement
AI FDEs do not start with the model. They start with the workflow.


05
What AI FDEs Actually Build
AI FDEs build systems, not isolated prompts
AI applications
RAG assistants
AI agents
Workflow copilots
Multi-agent systems
Domain-specific AI apps
Control & operations
Evaluation harnesses
Telemetry dashboards
Governance evidence packs
Human approval workflows
Delivery outputs
Production handover runbooks
Deployment-ready capability packs
Common thread
Every build must be usable, governed, and measurable.
The output of an AI FDE is not a demo. It is a usable, governed AI capability.


06
The Technical Spine Of AI FDE
AI FDEs need enough technical depth to make safe architecture decisions
1
User
2
AI App / Agent
3
Model
4
Retrieval / Tools
5
Enterprise Systems
6
Evaluation
7
Logs
8
Human Approval
Core technical concepts
LLM behaviour & prompting
RAG & grounding
Vector & hybrid search
Agent roles & tool use
API integration
RBAC & least privilege
Content safety & guardrails
Evals, traces & monitoring
Technical judgment means
Know the system boundary
Design safe integrations
Measure what matters
An AI FDE must know where the model ends and the system begins.


07
The Non-Technical Spine Of AI FDE
The hardest AI deployment problems are often human, organisational, and governance-related
Vague requirements
Stakeholder disagreement
Incomplete data
Pressure for speed
AI FDE in the operating environment
Security wants control
Legal wants evidence
Users need trust
Leadership wants measurable impact
The AI FDE must translate between
Business
Engineering
Security
Risk
Operations
End users
AI deployment fails when the operating environment is ignored.

08 The AI FDE Mindset

AI FDEs think in trade-offs, not tool excitement

Weak AI builder asks

"Can AI do this?"

Strong AI FDE asks

- Should AI touch this workflow?
- What evidence should it use?
- What must remain human-owned?
- What can go wrong?
- What needs to be logged and evaluated?
- How will value be proven?

Mindset shift

- From capability to consequence
- From output to outcome
- From speed alone to responsible acceleration

The AI FDE mindset is responsible acceleration.

09 Real-World AI FDE-Style Patterns

Across industries, the pattern is consistent: workflow first, AI second, governance always

Aviation
Predictive maintenance and aircraft uptime

Healthcare
Hospital command center and patient-flow operations

Maritime trade
Governed AI agent workforce

Banking
Regulated multi-agent service platform

Telecom
Production AI app portfolio

Life sciences
Enterprise GPT factory with governance

Different domains, same deployment pattern.

Different domains, same FDE pattern: discover deeply, build responsibly, govern continuously, prove value.


10
The AI FDE Readiness Bar
AI FDE readiness is proven oy field judgment, not tool familiarity
Real-world constraints
Unclear problem
Imperfect data
Probabilistic model
Political workflow
Governance friction
Security challenges
Need for user trust
Pressure for outcomes
Five readiness signals
1
Discovery Depth
Uncover the real workflow pain
2
Architecture Judgment
Choose the right AI pattern
3
Governance Discipline
Define controls, approvals, and risk boundaries
4
Evaluation Thinking
Measure quality, safety, latency, cost, and value
5
Stakeholder Defence
Explain trade-offs across business, engineering, security, legal, and operations
Ready when | Field judgment meets responsible execution
An AI FDE is ready when they can convert ambiguity into architecture, architecture into a working AI system, and the AI system into governed business value.

Case Studies

CS01

Airbus Skywise —
Aviation Operations &
Predictive Maintenance

CS02

Cleveland Clinic + Palantir —
Healthcare Virtual
Command Center

CS03

AD Ports Group —
AI Agents Workforce for
Maritime Trade

CS04

Banco Bradesco —
Multi-Agent Banking
Platform

CS05

Hughes / EchoStar —
Production AI App
Portfolio

CS06

Moderna —
Enterprise GPT Factory
in Life Sciences

OTECHADEMY

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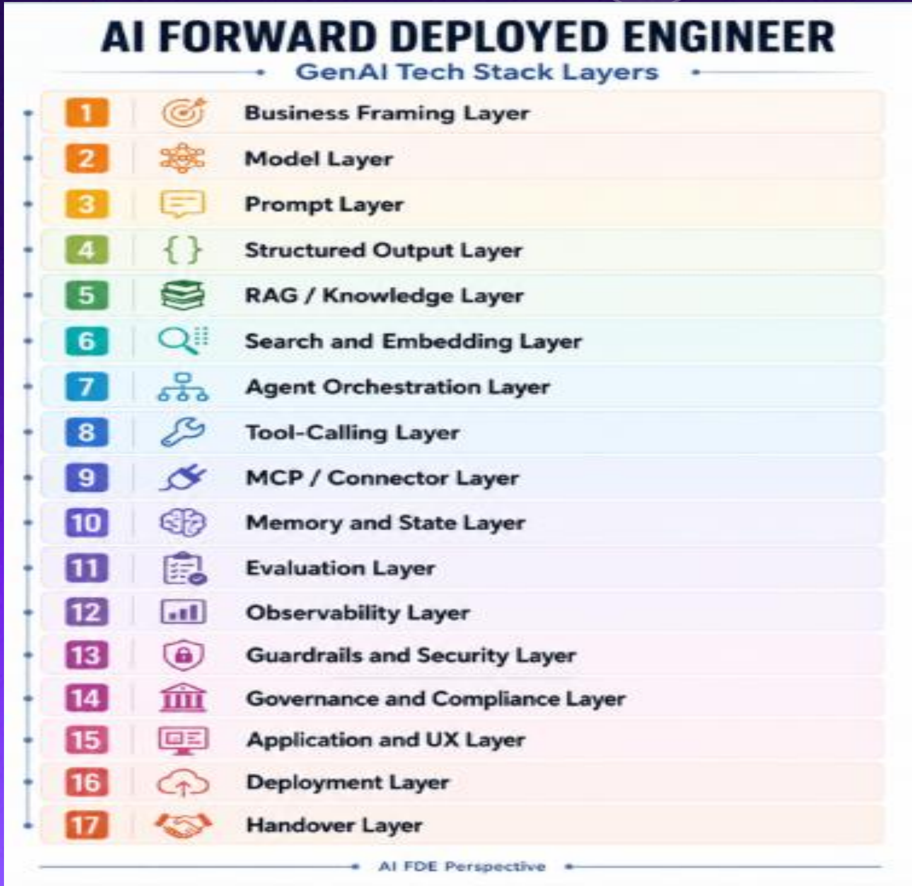

Layer		FDE meta-skill
1. Business Framing Layer	Business problem, workflow pain, stakeholder need, success metric, risk boundary, adoption path	Convert ambiguity into a scoped GenAI use case
2. Model Layer	LLMs, reasoning models, multimodal models, embedding models, small/local models, model selection criteria	Select the right model based on cost, latency, privacy, reasoning, context length, and enterprise fit
3. Prompt Layer	System prompts, prompt templates, few-shot examples, business rules, prompt versioning, prompt injection risks	Convert domain rules and SOPs into reliable model instructions
4. Structured Output Layer	JSON output, schemas, validation, error handling, retries, downstream application handoff	Make GenAI outputs usable by real applications and workflows
5. RAG / Knowledge Layer	Ingestion, parsing, chunking, embeddings, retrieval, reranking, citations, freshness, access control	Build grounded copilots over enterprise knowledge
6. Search and Embedding Layer	Semantic search, vector search, keyword search, hybrid search, metadata filtering, ranking	Choose the right retrieval strategy for the business problem
7. Agent Orchestration Layer	Workflows vs agents, planning, state, tool use, retries, handoffs, human approval	Build controlled agents that can act safely and recover from failure
8. Tool-Calling Layer	Function calling, API integration, tool schemas, permissions, side-effect control, tool logs	Connect GenAI safely to enterprise systems and business actions
9. MCP / Connector Layer	MCP hosts, clients, servers, resources, tools, prompts, consent, authorization	Understand how AI apps connect to external systems without assuming trust

10. Memory and State Layer	Session memory, long-term memory, task state, retention, deletion, memory poisoning risk	Decide what the system should remember, forget, and protect
11. Evaluation Layer	Golden datasets, RAG tests, agent tests, regression tests, rubric grading, red-team tests	Prove that the AI system works before production
12. Observability Layer	Prompt traces, retrieval traces, model traces, tool-call traces, latency, cost, failures	Debug the full AI workflow end to end
13. Guardrails and Security Layer	Prompt injection, data leakage, unsafe output, excessive agency, supply-chain risk, rate limits	Design blast-radius-limited GenAI systems
14. Governance and Compliance Layer	AI inventory, risk classification, data classification, impact assessment, audit logs, incident response	Make the AI system acceptable to business, legal, risk, compliance, and security teams
15. Application and UX Layer	Chat UI, workflow UI, approval screens, dashboards, citations, feedback capture	Convert the AI backend into a usable enterprise product
16. Deployment Layer	APIs, containers, CI/CD, cloud deployment, secrets, monitoring, scaling, cost control	Move from prototype to production safely
17. Handover Layer	Architecture docs, runbooks, model/system cards, evaluation report, risk register, operating guide	Leave behind a maintainable system, not a heroic prototype